Zichuan Yao

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SKILLS

Technical: Python, C/C++, MATLAB, R, TypeScript, Assembly; Git, GitHub, Linux, LaTeX, NextJS; PINN, Transformer, GFlowNet, NVIDIA PhysicsNeMo Sym(Pytorch Based), FEniCSx, PyBamm

WORK EXPERIENCE

University of Waterloo — Jan – Apr. 2026

Research Assistant — *Waterloo, ON*

- **PINN Development with NVIDIA PhysicsNeMo**: Built Physics-Informed Neural Networks via NVIDIA framework; designed custom sampling method and tailored MLP with adaptive step sizes, adaptive weighting, and selected appropriate activation functions
- **Multi-Task Learning for Coupled Systems**: Pre-trained on key physical equations then fine-tuned on interdependent components to enhance stability and efficiency for coupled equation systems.
- **Numerical Simulation & FEM/FVM Implementation**: Developed and validated computational models using FEniCSx and PyBamm, applied finite element/finite volume methods, and reproduced high-precision results from published papers.
- **Parametric Simulation & Data Processing**: Designed automated parameter studies, analyzed simulation outputs, and optimized numerical schemes for robust model performance.
- **Code Management & Documentation**: Maintained version control with Git/GitHub, documented computational workflows, and presented simulation results in group meetings.

Shanghai Jiao Tong University — Sept – Dec. 2025

Research Assistant — *Shanghai, China*

- **Crack Path Analysis Model Optimization & Visualization**: Migrated and refactored a Dijkstra-based crack-path analysis algorithm from MATLAB to Python, added step-wise visualization, and improved documentation for maintainability and clarity.
- **Crack Formation Model Development**: Designed a Python model linking crack-initiation density with oxide-film growth in zirconium alloys, converting literature-based mechanisms into an executable simulation tool for quantitative crack-formation studies.
- **Technical Documentation & Project Knowledge Base**: Authored clear technical documentation and user manuals for the crack-formation model and optimized analysis program, detailing model principles, function interfaces, and usage workflows to lower the technical barrier for future users.

Porton Pharma Solutions Ltd. — May – Aug. 2024

Data Analyst — *Shanghai, China*

- **Performed multi-source analyses** of global pharma/biotech companies, providing background checks and pipeline transaction insights for partnership decisions.
- **Analyzed drug dosage data** across preclinical to commercial stages, benchmarking similar drugs and supporting API-dosage forecasting for unmarketed products.
- **Researched the GLP-1 drug landscape** and produced a PowerPoint report outlining key growth drivers and market projections.
- **Managed weekly lead data**, ensuring accurate and timely Excel entry for effective lead tracking and follow-up.

PROJECTS

Personal Webs | Typescript, HTML, NextJS, Shadcn, Magic UI

- A responsive portfolio website that showcases my work, includes a project gallery, and features contact info.
- Website address

Iris Neural Network | Python, Numpy, Pandas

- Built a deep neural network from scratch using NumPy to classify Iris flower species. Implemented data preprocessing, He/Xavier initialization, activation functions, forward and backward propagation, and optimization.

EDUCATION

University of Waterloo — Sept. 2023 to May. 2028

Bachelor of Honours Computational Mathematics, Co-operative Education — *Waterloo, ON*